

RESEARCH INTERESTS

My research interest lies in aligning generative models with human intention through the principles of reinforcement learning and probabilistic inference. I enjoy applying aligning techniques to real-world application such as visual generation [1,4,5], robotics control [1,8], and biological sequence design [4,7,9]. Recently, I have been focusing on aligning a few-step generative model for physical agents and scientific discovery.

EDUCATION

3/2025 - Current	Ph.D Student in Industrial and Systems Engineering Supervised by Jinkyoo Park.	KAIST
3/2023 - 2/2025	M.S in Graduate School of Artificial Intelligence Supervised by Jinkyoo Park. Thesis: Generative Data Augmentation Method for Suboptimal Control Data	KAIST
3/2016 - 2/2022	B.S in Mechanical Engineering Including 2 years of military service	Sungkyunkwan University

PUBLICATIONS

*: Equal Contribution

[1] Aligning Few-Step Generative Models by Amortizing Sample-based Variational Inference <u>Jaewoo Lee</u> *, Hyeongyu Kang*, Dohyun Kim, Kyuil Sim, Woocheol Shin, Minsu Kim, Taeyoung Yun, Jeongjae Lee, Sanghyeok Choi, Tabitha Edith Lee, Jongchul Ye, Jinkyoo Park Link to paper/code	<i>SPIGM@ICML 26</i>
[2] Automated Kernel Discovery Towards Understanding High-dimensional Bayesian Optimization Taeyoung Yun, Woocheol Shin, Inhyuck Song, <u>Jaewoo Lee</u> , Jinkyoo Park Link to paper	<i>DEMO@ICML 26</i>
[3] Robust Exploration through Generative Replay Inhyuck Song, Inhyuck Song, Taeyoung Yun, <u>Jaewoo Lee</u> , Jinkyoo Park	<i>RLxF@ICML 26</i>
[4] Diffusion Alignment as Variational Expectation-Maximization <u>Jaewoo Lee</u> , Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, Taeyoung Yun, Kiyoung Om, Jinkyoo Park Link to paper/code	<i>ICLR 26</i>
[5] Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function <u>Jaewoo Lee</u> *, Hyeongyu Kang*, Woocheol Shin*, Kiyoung Om, Jinkyoo Park Link to paper/code	<i>ICLR 26</i>
[6] Adaptive Replay Buffer for Offline-to-Online Reinforcement Learning Chihyeon Song, <u>Jaewoo Lee</u> , Jinkyoo Park Link to paper	<i>AISTATS 26</i>
[7] Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization Taeyoung Yun*, Kiyoung Om*, Sujin Yun, <u>Jaewoo Lee</u> , Jinkyoo Park	<i>ICML 25</i> <i>FPI@ICLR 25</i>

[Link to paper/code](#)

[8] GTA: Generative Trajectory Augmentation with Guidance for Offline Reinforcement Learning

[Jaewoo Lee](#)*, Sujin Yun*, Taeyoung Yun, Jinkyoo Park

[Link to paper/code](#)

NeurIPS 24
Gen4DM@ICLR 24

[9] Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization

Taeyoung Yun, Sujin Yun, [Jaewoo Lee](#), Jinkyoo Park

[Link to paper/code](#)

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WORKING EXPERIENCES

3/2023 - Current	Cofounder & Research Engineer	MongooseAI
	Designing an LLM agent for analyzing manufacturing big data Developing a prognostic health management system for rotating machinery Project management of building MLOps system for semiconductor tools	
2/2022 - 2/2023	Software Engineer	Samsung Electronics, AI Center
	Administering database system for semiconductor manufacturing	
7/2021 - 8/2021	Intern Data Scientist	Samsung Electronics, AI Center
	Analyzing manufacturing log data for process lead time optimization	

HONORS & AWARDS

9/2021	Dean's list	Sungkyunkwan University
	Honor for academic excellence	
9/2021	Engineering Innovation Award	Sungkyunkwan University
	AI-driven remote control system for surgical robot	
2/2021	3rd place	Korea Shipbuilding and Offshore Engineering
	Competition on manufacturing process optimization with AI	
9/2020	Dean's list	Sungkyunkwan University
	Honor for academic excellence	

OTHER EXPERIENCE

3/2021 - 2/2022	Team Leader	Student Club of AI, Sungkyunkwan University
12/2020 - 2/2021	Intern Researcher	Biomedical Modeling & Control Lab, Seoul National University
7/2017 - 3/2019	Military Service	Republic of Korea Army